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MILITARY SPECIFICATION

LADDER, ALUMINUM, THREE-WAY COMBINATION: STEP, STRAIGHT, EXTENSION, GENERAL SPECIFICATION FOR

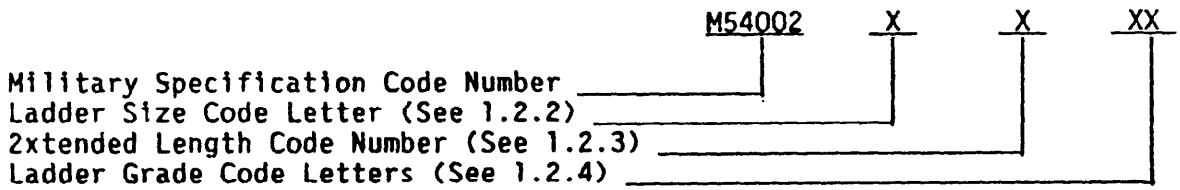
This military specification has been prepared by the Naval Air Systems Command based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-L-54002. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of three-way combination (step, straight and extension) aluminum ladders. As a step ladder it is adaptable to surfaces in different planes.

1.2 Classification. The ladders described herein are classified according to size, length and grade.

1.2.1 Part number. The military specification part number is a definitive part number which corresponds to the sizes, extended lengths and grades of ladders covered by this specification. The military specification number, and the size, length, and grade of ladder codes are combined to form a definitive part number.



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5440

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1.2.2 Ladder size. The ladder size is identified by a single letter (see Table I).

TABLE I. Ladder size code letter.

CODE LETTER	SIZE, FEET
A	6
B	8
C	10
D	11

1.2.3 Extended length. The ladder extended length is identified by a single digit number (see Table II).

TABLE II. Extended length code number.

CODE NUMBER	EXTENDED LENGTH (FEET)
1	10
2	14
3	18
4	19

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1.2.4 Ladder grade. The ladder grade is identified by two letters (see Table III).

TABLE III. Ladder grade code letters.

CODE LETTERS	GRADE
AA	Grade A, Heavy duty
BB	Grade B, Medium duty
CC	Grade C, Light duty
DD	Grade D, Extra heavy duty

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-S-781	Strapping, Steel and Seals
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-640	Box, Fiberboard, Corrugated, Triplewall
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MIL-T-704	Treatment and Painting of Materiel
MIL-G-20241	Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive

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MIL-P-23377	Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant
MIL-C-52950	Crate, Wood, Open and Covered
MIL-S-81733	Sealing and Coating Compound, Corrosion Inhibitive

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-889	Dissimilar Metals

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American National Standards Institutes (ANSI).

ANSI A14.2	Safety Requirements for Portable Metal Ladders
ANSI/ASME	Boiler and Pressure Vessel Code, Section IX
ANSI/AWS D1.1	Structural Welding Code, Section 5

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018).

American Society for Testing and Materials (ASTM).

ASTM D-3951	Standard Practice for Commercial Packaging

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3). The ladder shall conform to ANSI A14.2.

3.2 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.2.1 Material deterioration and control. The ladders shall be fabricated from compatible materials inherently corrosion resistant or, treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and/or operating environment (see 6.1) to which the item may be exposed.

3.2.1.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation shall be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241.

3.2.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or his designated representative.

3.3 Construction.

3.3.1 Front section. The front section shall consist of two side rails joined at equal intervals by steps. The front section shall have one step at its top which completely covers the ends of the side rails. The side rails shall be heavy gage heat-treated aluminum channels, minimum section 3 inches by 3/4 inch, and shall be the specified length plus or minus 1/2 inch. The steps shall be 15 inches in length minimum and 3 inches deep, and shall be

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spaced 12 inches plus or minus 1/8 inch center to center beginning at the center of the top step. All steps shall have a nonskid working surface and shall be parallel to each other and to the horizontal plane.

3.3.2 Back section. The two side rails of the back section shall be heavy gage heat treated aluminum alloy of suitable cross-section. The back section shall provide a means for leveling when the ladder is used as a step ladder in a stairway or other terraced or uneven surface. When fully extended, the back section shall become an extension of the front section and the combined length of the two sections shall be 10, 14, 18 and 19 feet respectively for the 6, 8, 10 and 11 foot step ladder size. The back section shall have rungs of suitable strength material, either round, square, or D-shaped, not less than 15 inches in length and shall be spaced 12 inches plus or minus 1/8 inch on center starting with the first step above the top step of the front section when the ladder is extended.

3.3.3 Leveling device. The minimum leveling device adjustable range for the back section shall be as shown below:

Step ladder size (feet)	Adjustable range (inches)
6	0 - 21
8	0 - 33
10	0 - 45
11	0 - 51

3.3.4 Step and rungs. There shall be no permanent deformation (see 4.5.2d) of the steps or rungs when tested in accordance with 4.5.2.1.

3.3.4.1 Bracing. The bottom step and every third step of the front section, and the first and last rung of the adjustable part of the back section shall be provided with bracing members. The bracing members shall form a triangle with the underside of the step or rung and the side rails when the ladder is in the extended position. The contractor's standard bracing design may be used subject to the approval of the contracting officer.

3.3.5 Side rails. There shall be no permanent deformation of the side rails when tested in accordance with 4.5.2.3.

3.3.6 Extension locking device. The extension locking device shall be designed to withstand all load tests.

3.3.7 Stops. Ladders shall be equipped in such a manner that the ladders, when used as extension ladders, shall not exceed the maximum working lengths specified.

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3.3.8 End caps. End caps shall be provided on the upper ends of each side rail of a combination section when the rail is not otherwise capped. End caps shall be securely attached.

3.4 Nonskid feet. The side rail ends of both front and back sections which rest on the surface shall be provided with nonskid pads. When erected as an extension, straight or step ladder the nonskid pads shall be in full contact with the surface. The dimensions of the nonskid pads shall be not less than the dimensions of the projected area outlined by the cross section of the end of the rail.

3.5 Spreaders. A spreader or locking device of sufficient size and strength to securely hold the front and back sections in the open position shall be a component part of each combination step-ladder, straight ladder and extension ladder. The spreaders shall not separate from the back and front sections with the ladder under load.

3.6 Slope. When the step ladder is erected on a level surface, the slope for the front and back section shall be 3-1/2 inches, and 2 inches plus or minus 1/4 inch respectively, for a foot length of side rail.

3.7 Performance.

3.7.1 Step ladder assembly. When erected on a level surface or on stairs, the step ladder shall withstand the tests in accordance with 4.5.2.2 without failure or deformation.

3.7.2 Stability of step ladder. When erected on a level surface or on stairs, the step ladder shall remain in a level position with all feet on the surface and shall withstand each pull force without raising the base of the side rails from the surface, when tested in accordance with 4.5.2.2.

3.7.3 Extension ladder assembly. The ladder in the extended position shall not fail or have any permanent set (see 4.5.2c) when tested in accordance with 4.5.2.3.1.

3.7.4 Erection, 75-1/2 degrees. When fully extended and erected at an angle of 75-1/2° to the horizontal, the ladder shall not fail or show any permanent set of the side rails when tested in accordance with 4.5.2.3.2.

3.7.5 Front rails. The front rails of the ladder shall show no visible defects (see 4.5.2a) or permanent deformation when tested in accordance with 4.5.2.4.

3.7.6 Rear rails. The rear rails of the ladder shall show no visible defects or permanent deformation when tested in accordance with 4.5.2.5.

3.7.7 Compression resistance. The ladders shall withstand, without failure or permanent deformation, the amount of compression specified for each grade when tested in accordance with 4.5.2.6.

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3.8 Identification marking. The ladders shall be identified in accordance with MIL-STD-130.

3.9 Treatment and painting. Unless otherwise specified (see 6.2), the ladders shall not be painted.

3.9.1 Painting. When painting is required the ladders shall be cleaned, treated and painted in accordance with MIL-T-704. Color shall be as specified (see 6.2).

3.10 Workmanship. All parts of the ladder, and welded parts shall be clean and free from sand, dirt, pits, scale, grease and other harmful, extraneous material that could be ingested into aircraft engines.

3.10.1 Welders and welding.

3.10.1.1 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications

ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.

3.10.1.2 Welding. The surface of parts to be welded shall be free from scale, paint, grease and other foreign matter. Welds shall transmit stress without permanent development or failure when the parts connected by the welds are subjected to proof and service loading. All welds shall be free from undercuts, cracks and closely spaced in-line surface porosity.

3.10.1.3 Aluminum welding. Aluminum welding shall be done by the metal-inert-gas method or any other method that will provide equivalent mechanical properties of the filler metal-to-base metal combinations. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed, or chipped but shall not be hammered. Preheating for welding is permissible provided the temperature does not exceed 400 F for a total of 30 minutes.

3.10.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. All bolts and nuts shall be correctly torqued.

3.10.3 Riveted connections. Rivets shall fill the holes completely. The upset rivet head shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the member. Only solid core rivets shall be used and shall go completely through the structural member.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. Inspection of packaging (see 4.6)

4.3 First article inspection.

4.3.1 Examination. The first article ladder shall be examined as specified in 4.5.1. Presence of one defect shall be cause for rejection.

4.3.2 Test. The first article ladder shall be tested as specified in 4.5.2 through 4.5.2.6. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Lot. All ladders of the same size and grade offered for inspection at one time shall constitute a lot.

4.4.2 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Level II.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.4 Tests. Samples selected in accordance with 4.4.2 shall be tested as specified in 4.5.2.1 through 4.5.2.6. AQL shall be 2.5 percent defective.

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4.5 Inspection procedures.

4.5.1 Examination. The ladders shall be examined as specified herein for the following defects:

101. Material not as specified (see 3.2).
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and/or operating environment (see 3.2.1 and 3.2.1.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.2.1.1).
104. Contractor does not have documentation available for identification of material, material finished or treatments (see 3.2.1.2).
105. Dimensions not as specified (see 3.3).
106. Leveling range not as specified (see 3.3.3).
107. Steps and rungs not as specified (see 3.3.4).
108. Side rails not as specified (see 3.3.5).
109. Shelf working area not as specified (see 3.3.6).
110. Stops not as specified (see 3.3.7).
111. End caps not as specified (see 3.3.8).
112. Nonskid pads not as specified (see 3.4).
113. Spreaders not as specified (see 3.5).
114. Slope of front and back sections not as specified (see 3.6).
115. Identification marking incorrect or missing (see 3.8).
116. Workmanship not as specified (see 3.10).

4.5.2 Tests. The terms visible damage, failure, permanent set and permanent deformation shall be defined as follows:

a. Visible damage: Damage clearly evident to the naked eye without recourse to external measuring or magnifying devices.

b. Failure: Damage in which the structure or component is buckled, twisted, torn or fractured.

c. Permanent set: Any set in excess of 1/1000 of the effective span of an extension ladder side rail, or any set in excess of 1/100 of the effective span of a step ladder step or an extension ladder rung.

d. Permanent deformation: Permanent set of the ladder structure which is clearly visible in accordance with the definition of visible damage.

4.5.2.1 Rungs and steps.

4.5.2.1.1 Rung and step bending. Apply one of the following loads as specified herein:

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Ladder grade	Test load, pounds
A	1,000
B	900
C	800
D	1,100

The load shall be applied over a 3-1/2 inch width on the center of the longest or bottom step. The test load shall also be applied to the longest like step without braces as well as all steps of different design or material specifications. Inability of the step or rung to withstand the load without failure or permanent set in excess of 1/100 of the length of the step measured between the inside web of the side rails shall constitute failure of this test.

4.5.2.1.2 Rung or step side rail shear strength. Apply the applicable test load as specified in 4.5.2.1.1 over a 3-1/2 inch width on the longest step with the least fastening, as near the side rail as possible while the ladder is in an open position. Remove the load. Any indication of failure in the fastening means attaching the step or rung to the side rail shall constitute failure of this test.

4.5.2.1.3 Rung and step torque. Apply a torque load in accordance with Table IV to a 3-1/2 inch wide area at the center of a rung or step. The torque shall be applied using a test bar, that will not damage the rung or step, with a 30 inch long moment arm. The torque shall be applied first in a clockwise then in a counterclockwise direction for ten alternate load cycles. Any visible relative motion between the rung or step and the side rail during application of the torque load shall constitute failure of this test.

TABLE IV. Rung Torque Test Loads.

GRADE	TORQUE TEST (inch-pounds)	TEST LOAD (pounds)
A	900	30
B	810	27
C	720	24
D	990	33

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4.5.2.2 Stability. Set the ladder in the open position on a level floor. Apply a uniformly distributed test load in accordance with Table V to the ladder top step. Subject the ladder to a horizontal pulling force applied at the geometric center of the step ladders top cap, at a distance of not more than 1/2 inch above its top surface, in accordance with Table V to test for front, side and rear stability. Then apply the horizontal pulling force to the side at the top step to test for side stability. Inability of the ladder to remain in a level position during testing or raising of the base of the side rails from the floor during testing shall constitute failure of this test.

TABLE V. Stability loads.

GRADE	UNIFORMLY DISTRIBUTED LOAD ON TOP STEP	PULLING FORCE (MINIMUM) (POUNDS)		
		FRONT	SIDE	REAR
A	250	25	20	45
B	225	25	20	45
C	200	25	20	45
D	350	50	25	50

4.5.2.3 Extension ladder.

4.5.2.3.1 Horizontal bending. Place the ladder in a flat, horizontal position, supported 6 inches from the ends of the side rails. Open the ladder to its maximum length as an extension ladder. In order to set the ladder, preload it to 75 percent of the test load as shown in Table VI and hold for one (1) minute and unload. Then apply a load as shown in Table VI equally to the side rails at the center of the ladder by means of a 3-1/2 inch wide beam. Evidence of permanent set in excess of 1/1000 of the effective span of the side rails or any other visible weakening of failure of the ladder shall constitute failure of this test.

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TABLE VI. Extension ladder horizontal bending loads.

GRADE	TEST LOAD (pounds)
A	250
B	225
C	200
D	350

4.5.2.3.2 Erection, 75-1/2 degrees. Set up the ladder in the extension position to its maximum length at the working angle of 75-1/2 degrees. A series of loads shall be applied to the step or rung in accordance with Table VII. Each load shall be applied to the step or rung at the center of the step or rung, over an area of 3-1/2 inches wide. Evidence of permanent set of the side rails or any other visible weakening or failure of the ladder shall constitute failure of this test.

TABLE VII. Location and loads.

EXTENSION LADDER SIZE (EXTENDED, FEET)	STEP AND RUNG NUMBER	GRADE OF LADDER AND LOAD PER STEP OR RUNG			
		A	B	C	D
10	3, 5, 7	250	225	200	350
14	5, 7, 9	250	225	200	350
18	7, 9, 11	250	225	200	350
19	8, 10, 12	250	225	200	350

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4.5.2.4 Front rail cantilever bending. Open the ladder and place on its side. Clamp the lower side rails to a horizontal support. Apply a load over a 2 inch wide block across the side rail parallel to the bottom edge in accordance with Table VIII. Remove the test load. Evidence of visible damage or permanent deformation to the front side rail section shall constitute failure of this test.

4.5.2.5 Rear rail cantilever bending. Open the ladder and place on its side with one set of rails clamped to a horizontal support. Apply the test load over a 2 inch wide block across the side rail parallel to the bottom edge in accordance with Table VIII. Remove the test load. Evidence of visible damage or permanent deformation shall constitute failure of this test.

TABLE VIII. Step ladder cantilever bending test.

GRADE	FRONT RAIL CANTILEVER TEST LOAD (POUNDS)	REAR RAIL CANTILEVER TEST LOAD (POUNDS)
A	200	125
B	150	100
C	100	75
D	250	250

4.5.2.6 Step ladder compression. Set up the ladder in the open position and apply a load, as shown in Table IX, uniformly distributed over the ladder top step. Evidence of failure or permanent deformation of the ladder shall constitute failure of this test.

TABLE IX. Step ladder test load.

GRADE	TEST LOAD (pounds)
A	1000
B	900
C	800
D	1100

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, Level II.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

117. Material, methods and containers not as specified. Each incorrect material, method or container shall constitute one defect.
118. Blocking, bracing, cushioning and tiedown not provided as specified.
119. Packing not as specified.
120. Strapping not as specified in MIL-STD-129, Level A.
121. Marking illegible, incorrect, incomplete or missing.

5. PACKING

5.1 Packing. Packing shall be level A, B or Commercial as specified (see 5.2).

5.1.1 Level A. Each ladder shall be collapsed and movable parts secured with tape conforming to PPP-T-97, type II, or soft annealed steel wire. Ladders of like size and description shall be packed in wood cleated plywood boxes conforming to PPP-B-601, overseas type, in quantities not to exceed the weight limitations of the box specification. Blocking, bracing, cushioning and tiedowns shall be provided to prevent movement and damage to the contents. The boxes shall be closed and strapped in accordance with appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B unless finish A strapping is specified (see 6.2).

5.1.2 Level B. Each ladder shall be collapsed and movable parts secured with tape conforming to PPP-T-97, type II or soft annealed steel wire. Ladders of like size and description shall be packed in fiberboard boxes conforming to PPP-B-640, class 2, or wood crates conforming to MIL-C-52950, style A, type II, in quantities not to exceed the weight limitations of the box or crate specification. Closure and strapping shall be in accordance with the appendix of the box or crate specification, as applicable.

5.1.3 Commercial. The ladders shall be packed in accordance with ASTM-D-3951.

5.2 Marking. Marking for levels A and B packing shall be in accordance with MIL-STD-129. Marking for commercial packing shall be in accordance with ASTM-D-3951.

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6. NOTES

6.1 Intended use.

6.1.1 Grade A. Heavy duty ladders are intended for industrial use such as in construction or industrial maintenance.

6.1.2 Grade B. Medium duty ladders are intended for light industrial or commercial use in offices or light construction work, such as prefabricated buildings and replacing or installing partitions.

6.1.3 Grade C. Light duty ladders are intended for household use, such as painting and minor repairs.

6.1.4 Grade D. Extra heavy duty ladders are intended for military aviation maintenance use.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Military specification part number required (see 1.2.1).
c. When a first article is required for inspection and approval and the number of units required (see 3.1).
d. When ladders require treatment or painting (see 3.9).
e. Color of paint when required (see 3.9.1).
f. Degree of packing required (see 5.1 and 6.4).
g. When finish A strapping is required (see 5.1.1).

6.3 First article. When a first article inspection is required, the item shall be examined and should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.4 Marking. Preservation requirements have not been included in this document. For the purpose of shipping, marking shall be designated as A/A, B/B or Commercial/Commercial.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

6.6 Subject term listing.

Aluminum
Aviation
Commercial
Extension
Heavy, duty
Industrial
Ladder
Light, duty

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Maintenance
Medium, duty
Repairs, minor
Step
Straight
Three-way, combination

6.7 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing Activity:
Navy - AS
(Project No. 5440-N010)

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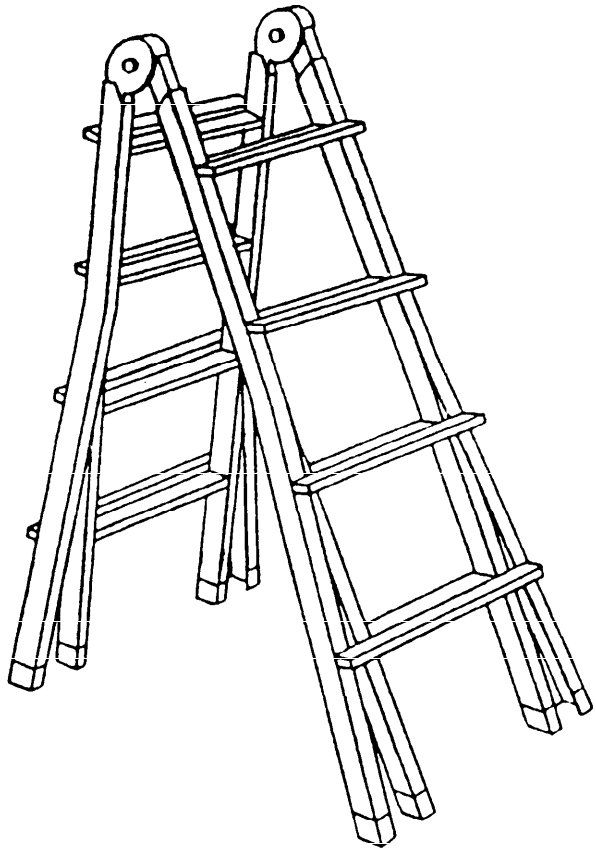


FIGURE 1. Step ladder configuration for M54002-D-4-DD.

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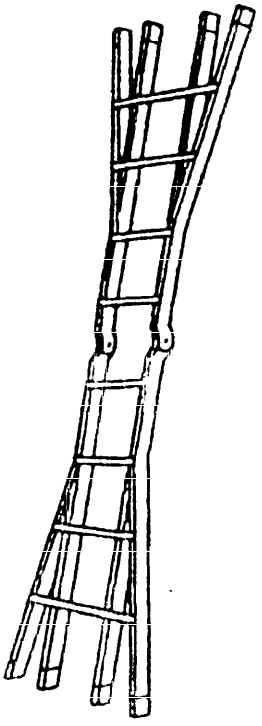


FIGURE 2. Straight ladder configuration for M54002-D-4-DD.

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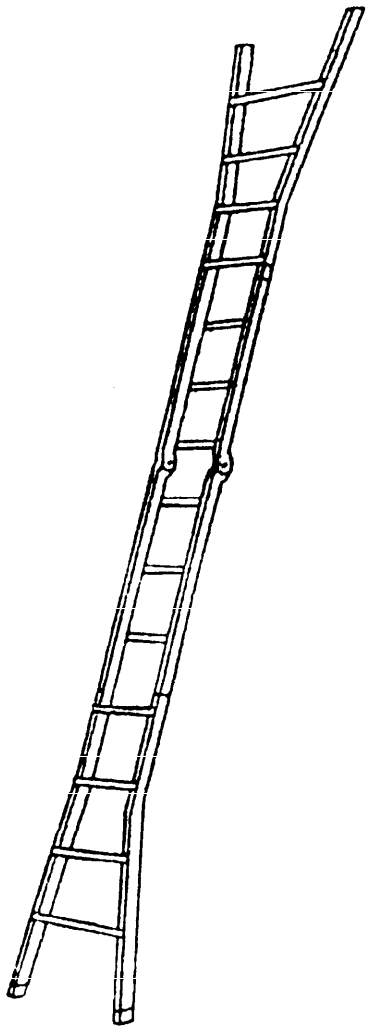


FIGURE 3. Extended ladder configuration for M54002-D-4-DD.

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